

COURSE DESCRIPTION

Course Number	0418 - 311	Course Title	Numerical Computation
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Hazem Raafat
Compulsory	Yes	Prerequisite	0410-102, 0410-111, 0418-221

Textbook

Title:	Python Programming and Numerical Methods: A Guide for Engineers and Scientists
Author:	Qingkai Kong, Timmy Siau, Alexandre Bayen
Publisher:	Academic Press, Dec. 2020.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction, computer limitations, and sources of errors	3
2	Introduction to Python	3
3	Linear and non-linear equations	6
4	Root finding algorithms using iterative methods	3
5	Matrix operations and solving systems of linear equations	3
6	Polynomial interpolation	3
7	Regression	3
8	Numerical differentiation	3
9	Numerical integration	6
10	Simple optimization algorithms	6
	Exams	3
	Total	42

Lab Schedule

	Topics	Hours
1	Python Installation, Python Introduction (input/output up to control statements) using mathematical problems (quadratic formula, slope, primality, factorization, geometry, etc.)	1
2	Advanced Python (Functions, Lists, tuples and Files), and introducing the NumPy Library	1
3	Linear and Non-Linear equations	1
4	Iterative methods (bisection, newton, etc.)	2
5	Matrix operations and solving systems of linear equations	2
6	Polynomial interpolation	1
7	Regression	1
8	Numerical differentiation	1
9	Numerical integration	2
10	Simple optimization algorithms	1
	Total	13

Grading

20%	HWs/labs/ project/quiz
20%	Midterm 1
20%	Midterm 2
40%	Final

Course Outcomes

Students should be able to:

1. Analyze key properties of numerical algorithms, such as error propagation analysis.
2. Understand and analyze common pitfalls in numerical computing.
3. Perform data analysis efficiently and accurately using data fitting methods.
4. Understand and analyze numerical methods for Integration, differentiation, interpolation.
5. Perform optimization using simple algorithms.
6. Implement a range of numerical algorithms efficiently using modern scientific computing programming languages.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M	M				
2	M	M				
3	M	L				L
4	M	L				L
5	L	M				M
6	H	H				H

Last updated on April 4, 2021 by Dr. Hazem Raafat

Approved by the curricula committee of the CS department on April 5, 2021